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A

B

C

D

E

F

Inputs

AFE

Digital

Power

12-lead portable ECG – 10 electrodes, 8 simultaneous 24-bit channels

45 x 45 mm, 4 layers; protected 1S cell; USB-C charging; BLE batches <=1 s target

Read DESIGN_SPEC.md and DESIGN_REPORT.md before prototype assembly.

Only connect electrodes when running from battery; USB/programming requires electrode removal.

Battery-only patient connection; USB/SWD require electrode disconnection

Engineering prototype

Sheet: /

File: ECG12_Portable2.kicad_sch

Title: ECG12 Portable – System overview

Size: A3

Date: 2026-09-07

Rev: B

KiCad E.D.A. 10.0.1

Id: 1/5

File: Inputs.kicad_sch

File: AFE.kicad_sch

File: Digital.kicad_sch

File: Power.kicad_sch

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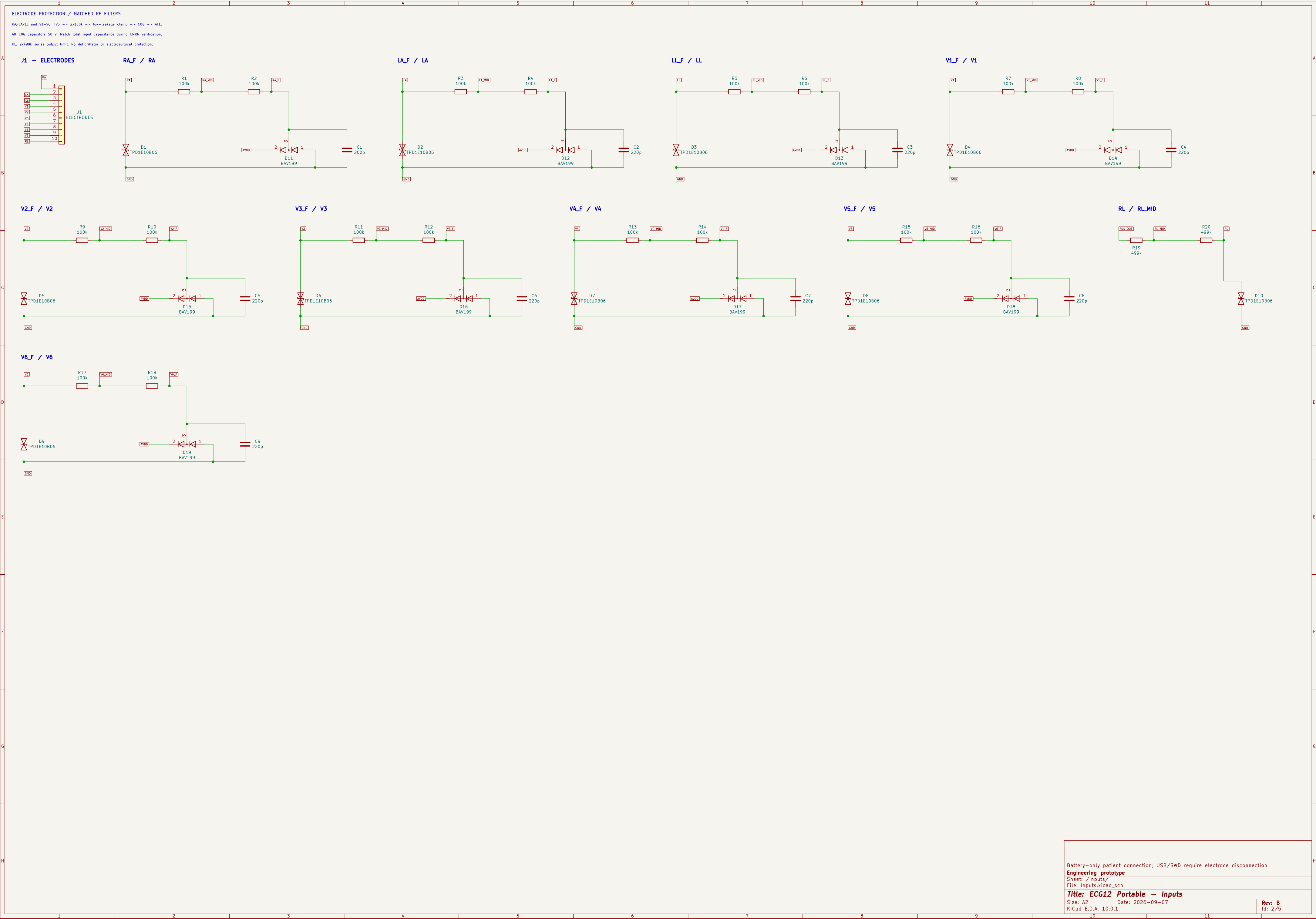
4

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8-CHANNEL SYNCHRONOUS AFE / RLD / WILSON TERMINAL

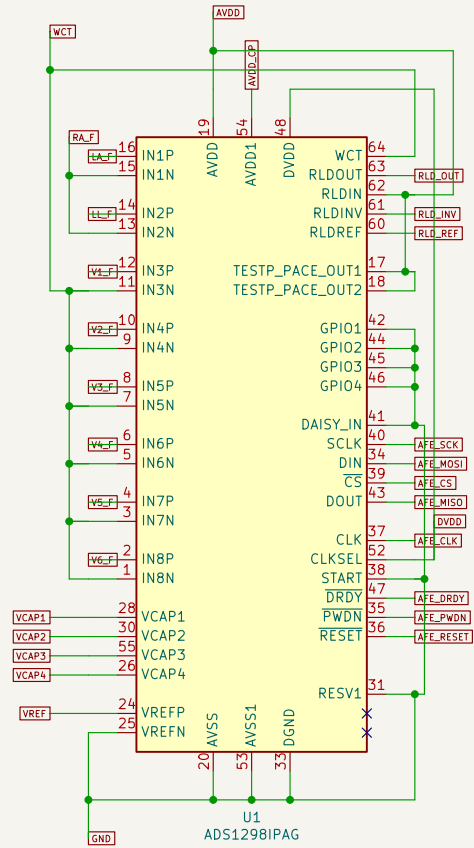
CH1=LA-RA, CH2=LL-RA; CH3-8=V1-V6 minus WCT.

WCT1=0x09, WCT2=0xC2; RA=CH1N, LA=CH1P, LL=CH2P.

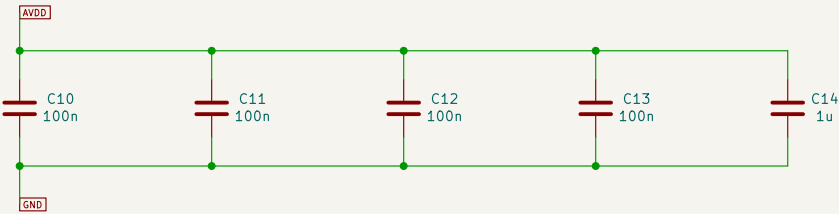
START pin low; use SPI START/STOP commands. Keep unused GPIO inputs.

Internal VREF=2.4 V, GAIN=6; 500 SPS monitoring (sinc response is not flat to 150 Hz).

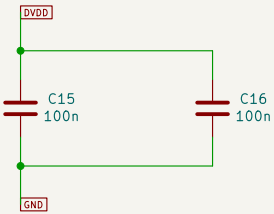
Synchronous ECG converter



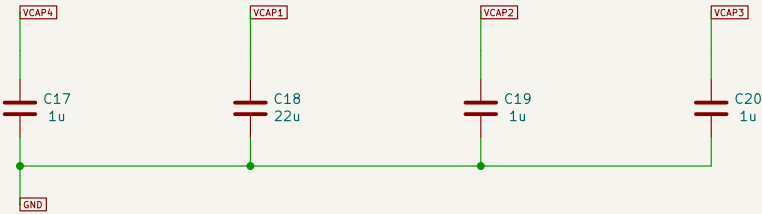
Analog supply bypass bank



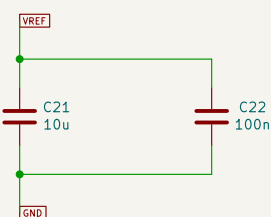
Digital supply bypass bank



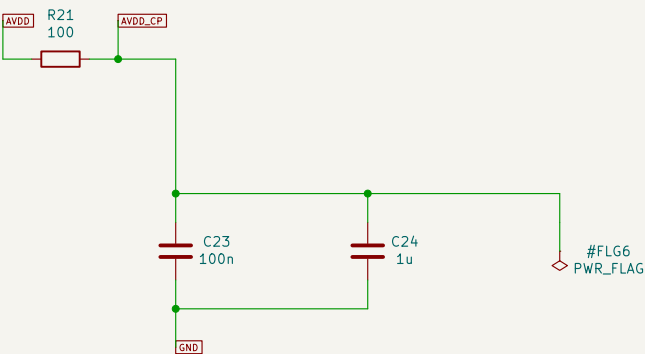
Internal bypass nodes



Internal 2.4V reference



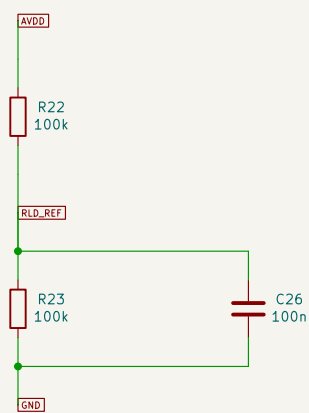
Charge pump supply filter



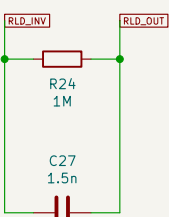
Wilson central terminal



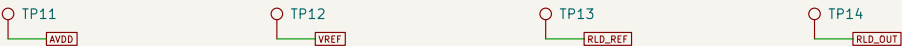
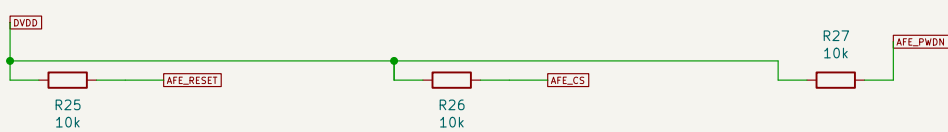
Patient bias reference



RLD compensated feedback



AFE control defaults



Battery-only patient connection; USB/SWD require electrode disconnection

Engineering prototype

Sheet: /AFE/

File: AFE.kicad_sch

Title: ECG12 Portable - AFE

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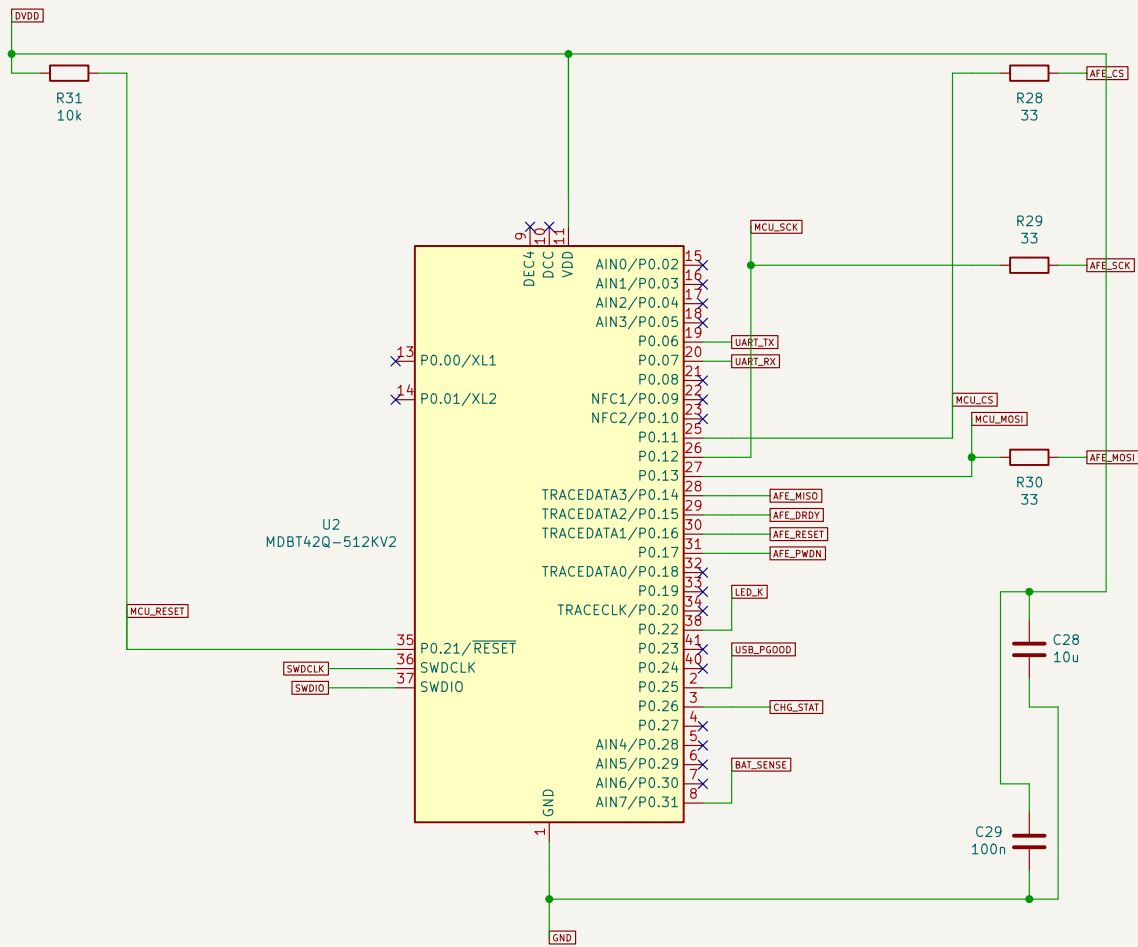
BLE OPEN-CPU MODULE / BUFFERING / TEST

Firmware: LDO mode, calibrated LFRC; no external DCC/DEC4 network.

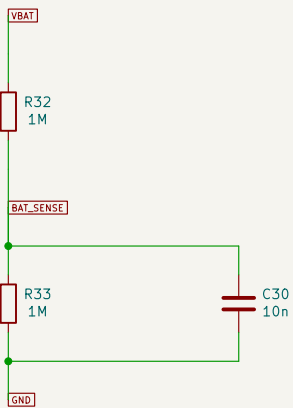
32 KiB RAM ring, batch every 250 ms; measured payload target >=200 kbit/s.

Maximum 1 s data-age target requires compatible central and bounded radio retries.

BLE module and SPI termination



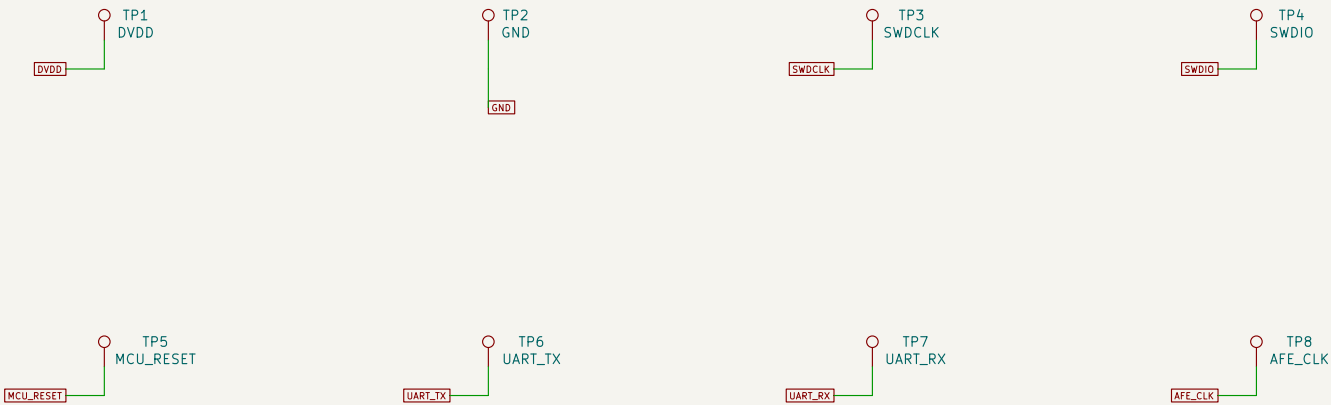
Battery voltage monitor



Status LED



Programming and test contacts



Battery-only patient connection; USB/SWD require electrode disconnection

Engineering prototype

Sheet: /Digital/
File: Digital.kicad_sch

Title: ECG12 Portable – Digital

Size: A3	Date: 2026-09-07	Rev: B
KiCad E.D.A. 10.0.1		Id: 4/5

